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1.0 Introduction

For this assignment, we are tasked to collect and analyse the dataset chose to identify the potential business insights. We collected a dataset that represents the customer's shopping behaviour in the fashion industry [1]. It provides comprehensive information about customer's product preferences and spending habits during their shopping experience. This dataset encompasses a range of variables, from customers profiling and clothing categories to payment details. We hope that through this dataset, we can clearly showcase the customers' shopping preferences with data visualization, thus aiding the company in identifying the market trends to increase the business sales and profits.

2.0 Dataset Overview

The dataset mainly focuses on the customers' spending behaviour in the fashion industry. The dataset is sourced from Kaggle, a data science platform which contains various types of datasets. It was first published by Shahzad Aslam and the dataset is scheduled to be updated annually. This dataset contains 3900 rows and 21 columns. The dataset is fully completed, and no null or duplicate data are detected. Below is the list of data that are included in this dataset:

1. Customer ID	6. Purchase Amount (USD)	11. Review Rating	16. Previous Purchases
2. Age	7. Location	12. Subscription Status	17. Payment Method
3. Gender	8. Size	13. Shipping Type	18. Frequency of Purchases
4. Item Purchased	9. Colour	14. Discount Applied	19. Discount Percentage
5. Category	10. Season	15. Promo Code Used	20. Customer Income Level
			21. Item Price

Table 1.0: Data Set List

3.0 Data Characteristics

Customer ID: A unique identifier assigned to each individual customer, facilitating tracking and analysis of their shopping behaviour over time.

Age: The age of the customer, providing demographic information for segmentation and targeted marketing strategies.

Gender: The gender identification of the customer, a key demographic variable influencing product preferences and purchasing patterns.

Item Purchased: The specific product or item selected by the customer during the transaction.

Category: The broad classification or group to which the purchased item belongs (e.g., clothing, electronics, groceries).

Purchase Amount (USD): The monetary value of the transaction, denoted in United States Dollars (USD), indicates the cost of the purchased items.

Location: The geographical location where the purchase was made, offering insights into regional preferences and market trends.

Size: The size specification of the purchased item, relevant for apparel, footwear, and certain consumer goods.

Color: The color variant or choice associated with the purchased item, influencing customer preferences and product availability.

Season: The seasonal relevance of the purchased item (e.g., spring, summer, fall, winter), impacting inventory management and marketing strategies.

Review Rating: A numerical or qualitative assessment provided by the customer regarding their satisfaction with the purchased item.

Subscription Status: Indicates whether the customer has opted for a subscription service, offering insights into their level of loyalty and potential for recurring revenue.

Shipping Type: Specifies the method used to deliver the purchased item (e.g., standard shipping, express delivery), influencing delivery times and costs.

Discount Applied: Indicates if any promotional discounts were applied to the purchase, shedding light on price sensitivity and promotion effectiveness.

Promo Code Used: Notes whether a promotional code or coupon was utilized during the transaction, aiding in the evaluation of marketing campaign success.

Previous Purchases: Provides information on the number or frequency of prior purchases made by the customer, contributing to customer segmentation and retention strategies.

Payment Method: Specifies the mode of payment employed by the customer (e.g., credit card, cash), offering insights into preferred payment options.

Frequency of Purchases: Indicates how often the customer engages in purchasing activities, a critical metric for assessing customer loyalty and lifetime value.

Discount Percentage: Indicates the discounted offer percentage applied when customer bought the item.

Customer Income Level: An ordinal scale qualitative data that shows the customer relative income position (e.g., low, medium, high).

4.0 Business Insights

4.1 Gender Spending Patterns

Gender plays a significant role in shaping customer purchasing behaviour, and the dataset provides valuable insight into how different genders engage with the platform. Through the analysis of the "Gender" and "Item Purchased" columns, we can identify product categories that are more likely to attract a particular gender.

For instance, the data reveals that a higher percentage of female customers tend to purchase apparel items, with roughly 65% of all clothing-related transactions attributed to women. On the other hand, categories like electronics, gadgets, and tech accessories seem to attract a larger male demographic, with approximately 55% of these transactions coming from male customers.

This gender-based preference helps businesses tailor their marketing campaigns and product placements. For example, targeted ads for clothing and accessories can be directed more toward female customers, while tech products and gadgets can be marketed with a focus on male customers. Understanding these patterns enables companies to optimize inventory and product development based on gender-specific preferences, ensuring that they meet the needs of their primary customer groups.

Furthermore, promotions and sales can be customized. For example, running gender-specific promotional offers such as “Women's Apparel Sale” or “Tech Gadgets for Men” could drive higher engagement and conversion rates. Thus, analysing gender-driven purchasing trends allows businesses to effectively position their products and increase their chances of attracting the right customer segment.

4.2 Age Group Dominance

Age is another crucial demographic factor that influences purchasing behaviour. The dataset's "Age" column offers an insightful view into how different age groups engage with online shopping. By analysing the data, we can observe distinct trends in how age impacts customer purchasing behaviour and item selection.

For example, a large proportion of customers aged between 18-34 years tend to purchase fashion items, accounting for roughly 40% of all transactions in this category. This age group is highly active in purchasing apparel and accessories, often driven by trends and the desire for new styles. In contrast, the 35-50 age group shows more interest in purchasing home appliances and electronics, contributing to around 30% of the total sales in those categories.

Interestingly, the 50+ age group exhibits a lower frequency of purchases overall, but when they do shop, their focus tends to be on household items and health-related products. This age group is responsible for approximately 15% of total sales in these categories.

By segmenting the customer base by age, businesses can better understand which products are popular among specific age groups. Targeted email campaigns, personalized product recommendations, and age-specific promotions could all be implemented to enhance customer engagement. For instance, the younger age group could be targeted with fashion and trend-driven content, while middle-aged customers might prefer more information on home appliances or technology gadgets.

4.3 Popular Categories and Items

The analysis of the "Category" and "Item Purchased" data provides valuable insights to customer purchasing behaviors. By identifying the most popular purchased categories, business can align their inventory, marketing efforts, and promotional strategies for the products that customers are most interested in.

From the dataset, Clothing and Accessories category contribute to the largest proportion of purchases, with Clothing contribute for 44.54% and Accessories for 31.79% of all transactions. Among the clothing items, the top-selling products include Blouses, Pants, Shirts, and Dresses, each contribute around 9-10% in the clothing category. While, in Accessories, Jewelry, Sunglasses, and Belts are the most popular items, each contribute around 13% in the accessories category.

Footwear represents another important category, contribute up to 15.36% of total purchases, followed by Outerwear at 8.31%.

This breakdown of categories shows how fashion products dominate the customer preferences. Seasonal trends highly influence the sales patterns, with items like socks, scarves,

and outerwear potentially seeing peaks during cold months, but the current data does not show a month-by-month breakdown.

These insights can help businesses plan their product inventory and promotional campaigns effectively. Since the clothing and accessories have top sales, companies can prioritize marketing in these areas. Seasonal products like sweaters and scarves can be promoted ahead of the winter season, while bundling complementary accessories like handbags and jewellery could enhance cross-selling opportunities.

4.4 Delivery Type Preferences

The dataset provides meaningful insights into how delivery type preferences vary among different customer segments. By analyzing the “Age”, “Gender”, and “Shipping Method” column, we can find the customer expectations for the speed, convenience, and cost.

Overall, Standard shipping remains the main choice, accounting for nearly 59% of all transactions. This preference suggests that a significant portion of customers prioritize affordability over speed when it comes to delivery. However, faster options like Express and 2-Day Shipping show strong appeal among younger customers, particularly those in the 18-25 and 26-35 age groups. For example, over 40% of shoppers aged 18–25 preferred Express shipping, while nearly 49% of those aged 26–35 chose 2-Day Shipping. This indicates that younger consumers are more willing to pay for quicker delivery, possibly due to lifestyle needs.

Interestingly, Store Pickup has become a choice among older customers, especially for those who are aged 36+, where it accounts for a notable percentage of shipping methods. Customers aged 51 and above highly prefer Standard and Store Pickup, show that convenience and lower costs is more important than speed as purchasing priorities in older customers.

To better meet customer expectations, businesses could offer flexible delivery options tailored by age and gender segments. For instance, emphasizing same-day or 2-day delivery promotions for younger audiences could boost sales, while offering free Standard shipping or convenient Store Pickup services may help retain older customers. These insights underline the importance of a segmented delivery strategy to optimize both customer satisfaction and operational efficiency.

4.5 Payment Method Preferences

Knowing how customers prefer to pay is key to making checkout fast, easy, and hassle-free. Looking at the data, we can see that payment preferences are pretty evenly spread out. Around 17% of customers use PayPal, Credit Cards, or Cash, while Debit Cards and Venmo are close behind, each with around 16%. Bank Transfers are slightly less popular, making up about 15.7% of transactions.

Even though no method completely dominates, digital wallets like PayPal and Venmo together represent a big slice of the pie — and they're growing. It's also interesting to see that Venmo users tend to spend the most, with an average transaction of about \$91, compared to around \$88–\$90 for other methods.

This suggests that customers using digital wallets may feel more comfortable making bigger purchases, possibly because they trust the convenience and security these options offer.

For businesses, offering multiple payment methods — especially digital wallets — could really make a difference. It not only gives customers more choice but could also encourage higher spending. Rewarding digital wallet users with loyalty perks or discounts might be a smart way to boost repeat purchases too.

4.6 Impact of Customer Income Level on Purchasing Behaviour

Looking at income levels gives us another important piece of the puzzle when it comes to understanding customers.

In this dataset, medium-income customers make up the largest group at about 35%, followed closely by high-income customers at 33%, and low-income customers at around 32%.

High-income shoppers tend to spend the most on average — around \$88 per transaction — often choosing more premium or luxury items. Medium-income customers spend a little less, about \$88 as well (very close!), focusing more on getting good value for their money.

Interestingly, the low-income group actually has a slightly higher average purchase at around \$90, which suggests they might be taking full advantage of deals, discounts, or saving up for specific purchases they really value.

These patterns show that businesses should think about tailoring their offers. For high-income shoppers, exclusive or luxury products could be a hit. For medium-income customers,

highlighting a balance of quality and price makes sense. And for low-income customers, continuing to offer great discounts and promotions would likely keep them coming back.

5.0 Conclusion

Analysing the shopping behaviour dataset has given us valuable insights into how different customers interact with products and promotions. It's clear that factors like gender, age, and income level all have a strong influence on what people buy, how much they spend, and even how often they return to shop again. For example, women and customers aged 18 to 34 play a huge role in driving fashion sales, showing that businesses should focus on trend-driven marketing to catch their interest.

The data also highlights the importance of knowing what products are most popular. Clothing, accessories, and electronics stand out as key categories, and businesses that stock and promote these items at the right time — like swimwear during summer — can really boost their sales. Discounts and promo codes turned out to be powerful tools too, not just in attracting customers, but also in keeping them loyal and coming back for more.

When it comes to payment methods, more people are turning to digital wallets, hinting that offering flexible and convenient payment options is no longer just a bonus — it's a necessity. And looking at customer income levels gives businesses even more opportunities to personalize their approach, whether it's offering premium, luxury products to high-income customers or focusing on promotions and value deals for budget-conscious shoppers.

Overall, this assignment shows how much businesses can learn just by looking closely at their data. Turning numbers into stories about customers' habits, needs, and preferences helps companies make smarter decisions, build stronger connections with their customers, and grow in a more meaningful way.

6.0 References

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